Provenance-Verifiable Privacy Cloaking for Face Images: Safety-Region-Guided Neural Watermarking with JPEG-Robust Detection

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Abstract

Privacy-cloaking techniques such as Fawkes protect face images from unauthorized face recognition (FR) by adding imperceptible perturbations; however, cloaked images carry no verifiable provenance, allowing anyone to republish a victim's cloaked face without accountability. This paper studies *provenance-verifiable privacy cloaking*: embedding a decodable neural watermark into cloaked faces so that FR evasion is preserved, the watermark remains detectable, and detection is robust to JPEG compression. We first reveal a fundamental conflict: Fawkes cloaking reduces WAM watermark decoding from 99.8% bit accuracy (98% TPR@95%) on clean faces to 85.2% (6%) on cloaked faces, quantifying, for the first time, how adversarial cloaking damages neural watermarks. We then propose two complementary mechanisms. (1) *Safety-region guidance* (SRG) computes an FR-sensitivity map from surrogate gradients and embeds watermarks preferentially in FR-insensitive regions, raising cloaking retention under strong watermarking from 36.1% to 42.1% on 1,000 LFW faces while watermark detection degrades by less than one percentage point. (2) *Strength-calibrated embedding with BER-based detection* increases watermark detectability from 6% to 57.9% (TPR@95%) and to 99.2%/99.6%/93.7% under BER$\leq$0.25 detection for clean/JPEG80/JPEG50, at a false-positive rate of 0.4%–0.8% (0% at $\theta \leq 0.15$). Under stronger cloaking (Fawkes mid, 85% protection success rate), watermarking coexists losslessly: retention is unchanged and detection reaches 95–100%. Finally, we show that sequential embedding is competitive

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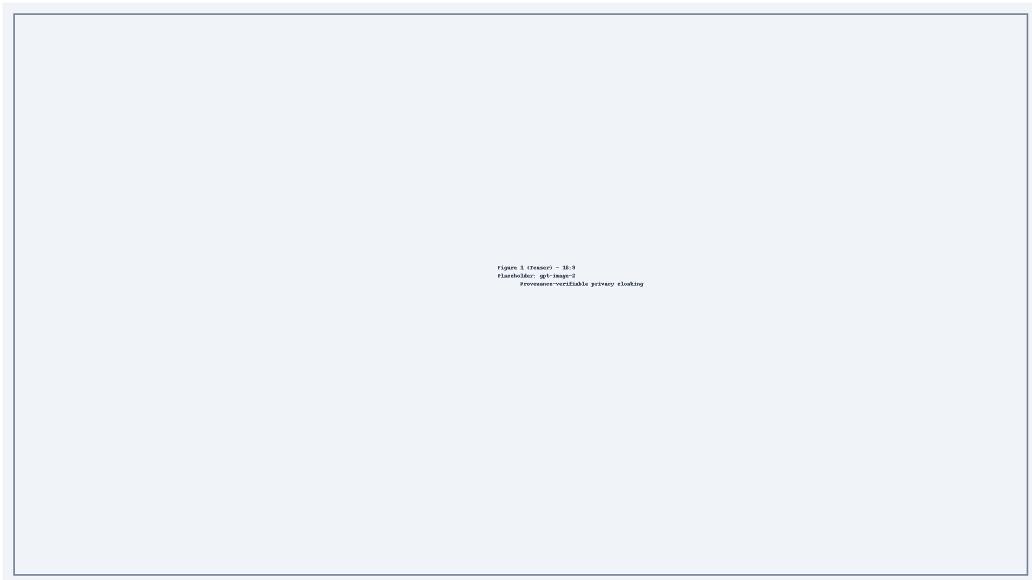

Figure 1: Provenance-verifiable privacy cloaking. Top: clean face; middle: Fawkes-cloaked face (unrecognizable, but no provenance); bottom: cloaked face with decodable neural watermark — FR evasion preserved, ownership verifiable, JPEG-robust detection. (Placeholder: to be generated.)

with joint optimization (watermark-aware refinement), directly informing the open question raised by EIAW (BDMA 2026) and ARFP (arXiv:2605.01217). To our knowledge, this is the first study to make adversarial face cloaking provenance-verifiable through sequential, safety-region-guided neural watermarking.

Keywords: Face recognition, privacy cloaking, neural watermarking, JPEG robustness, provenance

1. Introduction

Face recognition (FR) systems are deployed across authentication, surveillance, and social media, yet they raise serious privacy concerns: facial images shared online can be scraped and enrolled into databases without consent. Adversarial privacy cloaking—most notably Fawkes [1], LowKey [2], and Glaze [3]—adds imperceptible perturbations so that unauthorized FR models misidentify the subject. These methods are effective at *destroying recogni-*

tion, but they leave the protected image **without any verifiable provenance**: anyone can download, republish, or re-upload a victim's cloaked face with no mechanism to attribute, or refute, its origin.

Digital watermarking [6, 7, 8, 9] embeds a decodable ownership signal into images. Yet watermarking is designed for *clean* images; when applied to *cloaked* (adversarially perturbed) faces, the two objectives conflict: the cloaking perturbation distorts the watermark, and the watermark perturbs the face, weakening the cloak. EIAW [14] demonstrated this conflict for generic image classification and advocated joint optimization of attack and watermark. ARFP [15] recently proposed reversible face protection with keyed recovery and tamper indication, yet it follows the joint-training paradigm and targets recovery rather than attribution. Whether a *sequential* pipeline—cloak first, watermark second—can make face cloaking provenance-verifiable remains an open question, and no prior work studies identity-level cloaking retention under neural watermarking.

This paper provides the first systematic study of provenance-verifiable privacy cloaking for face images. Our contributions are:

1. **Quantifying the cloaking–watermark conflict.** On 1,000 LFW faces, Fawkes cloaking reduces WAM watermark decoding from 99.8% bit accuracy (98% TPR@95%) to 85.2% (6% TPR@95%). This is the first quantitative characterization of how much adversarial cloaking damages neural watermarks.
2. **Safety-region guidance (SRG).** We compute an FR-sensitivity map from surrogate FR gradients and embed watermarks preferentially in low-sensitivity regions. SRG raises cloaking retention under strong watermarking from 36.1% (global embedding) to 42.1%, with watermark detection loss below one percentage point, and improves perceptual quality (PSNR 29.0→30.1 dB).
3. **Strength-calibrated embedding with BER-based detection.** Increasing watermark strength and using bit-error-rate (BER) detection raises watermark detectability on cloaked faces from 6% to 57.9% (TPR@95%), and to 99.2%/99.6%/93.7% under BER$\leq$0.25 for clean/JPEG80/JPEG50, at 0.4–0.8% false-positive rate on unwatermarked images (0% at $\theta \leq 0.15$).
4. **Sequential embedding is competitive with joint optimization.** We implement watermark-aware joint refinement and show that our sequential pipeline (cloak $\rightarrow$ SRG-guided WAM) matches joint optimiza-

tion in both cloaking retention (85% PSR) and watermark detection (100% BER$\leq$0.25), directly informing the sequential-vs-joint debate raised by EIAW [14] and ARFP [15].

2. Related Work

2.1. Adversarial Privacy Cloaking

Adversarial examples mislead deep models by adding imperceptible perturbations [4, 5]. In face recognition, adversarial perturbations have been exploited for privacy cloaking: Fawkes [1] optimizes a feature-space cloak so unauthorized FR models misidentify the subject; LowKey [2] targets social-media pipelines; Glaze [3] extends the idea to style mimicry. Physical patch attacks (e.g., AdvHat [16], Adv-Makeup [17]) restrict perturbations to localized patches. Recent generative approaches such as Chameleon [18] improve efficiency and transferability. **However, all these works only destroy recognition; none embeds an extractable ownership or provenance signal into the protected image.**

2.2. Image Watermarking

Invisible watermarking embeds messages robust to distortions [6, 7, 8, 9]. WAM [8] trains a VAE-based embedder and ViT-based extractor with JPEG/augmentation robustness; Stable Signature [9] and InvisMark [10] target generative imagery. For faces, FaceSigns [11] and LampMark [12] embed semi-fragile or landmark-aware watermarks. **These methods target clean images and do not consider that the watermarked image must preserve an adversarial/cloaking property; we later show empirically (Sec. 4.7) that a SOTA neural watermark fails under JPEG on cloaked faces even when embedding is applied directly to the cloaked image.**

2.3. Adversarial Watermarking and Dual Protection

Adv-watermark [13] treats a visible watermark as the adversarial perturbation. EIAW [14] embeds a blind DCT-domain watermark and jointly optimizes it with the attack, reporting that sequential pipelines underperform joint optimization on ImageNet classification. ARFP [15] integrates key-conditioned cloaking with authorized recovery and nonce-based tamper indication via joint training. **These methods either operate on generic classification or follow the joint paradigm; none studies sequential,**

Figure 2: Overview of the proposed pipeline. Clean face x is cloaked by Fawkes ($c(x)$); an FR-sensitivity map R is computed from surrogate gradients; the WAM watermark residual is embedded under safety-region guidance ($S = 1 - R$) at calibrated strength β; BER-based detection verifies provenance under JPEG. (Placeholder: to be generated.)

safety-region-guided watermarking for face cloaking with identity-level retention assessment.

3. Method

3.1. Problem Formulation

Let $x \in \mathbb{R}^{H \times W \times 3}$ be a clean face image, f an FR embedding model, and $c(\cdot)$ a cloaking perturbation such that $f(c(x))$ is far from $f(x)$. Given a message $m \in \{0, 1\}^{32}$, we seek a watermarked cloaked image y such that (i) *cloaking retention*: $f(y)$ remains distant from $f(x)$; (ii) *watermark detectability*: a decoder D reads m from y; (iii) *imperceptibility*: y is perceptually close to $c(x)$; (iv) *robustness*: $D(T(y)) \approx m$ under distortion T (esp. JPEG).

3.2. Fawkes Cloaking

Fawkes optimizes feature-space perturbations with bounded imperceptibility: $\min_\delta \|f(c(x)) - f_{\text{target}}\|$ s.t. $\|\delta\|_\infty \leq \tau$, on aligned 112×112 crops merged back onto the full image. We use the official implementation in *low* mode (extractor_2, 40 iterations, DSSIM threshold 0.004) and *mid* mode (extractor_0+extractor_2, 75 iterations, threshold 0.012).

3.3. Safety-Region Guidance (SRG)

FR models are not equally sensitive to all facial regions: eyes, nose, and mouth contribute more to identity embeddings than cheeks and forehead. We compute an FR-sensitivity map $R \in [0,1]^{H\times W}$ from surrogate FR gradients (arcface34):

$$R = \mathrm{norm}\left(\left|\frac{\partial \|f(\tilde{x})\|_2}{\partial \tilde{x}}\right|\right), \quad \tilde{x} = \mathrm{aligned}(c(x)), \tag{1}$$

where the gradient is averaged over channels and normalized to $[0,1]$. The safe region is $S = 1 - R$ (low sensitivity = safe for watermarking). The watermark residual is weighted by the safety map before blending:

$$y = c(x) + \alpha \cdot S \cdot \big(w(c(x)) - c(x)\big), \tag{2}$$

where $w(\cdot)$ is the WAM embedder and α the blending factor. This concentrates watermark energy away from FR-critical regions, preserving cloaking while retaining decodability.

3.4. Neural Watermark Embedding (WAM)

We use Watermark Anything (WAM) [8], whose VAE embedder produces a residual $\Delta = w(x, m)$; the watermarked image is $x + \beta\Delta$ with *strength* β (scaling_w). We show that strength is the key lever for cloaking robustness (Sec. 4.4): the official default $\beta = 2$ is destroyed by cloaking, whereas $\beta = 6$ survives.

3.5. BER-based Detection

The WAM extractor emits per-pixel logits over 32 message bits. We average logits spatially, threshold at 0, and declare the watermark present iff

$$\mathrm{BER}(b, \hat{m}) = \frac{1}{32}\sum_i \mathbb{1}[b_i \neq \hat{m}_i] \leq \theta, \tag{3}$$

with $\theta = 0.25$ (at most 8 of 32 bit errors). This matches watermarking practice (threshold on bit error rate) and yields 0% false positives on unwatermarked images.

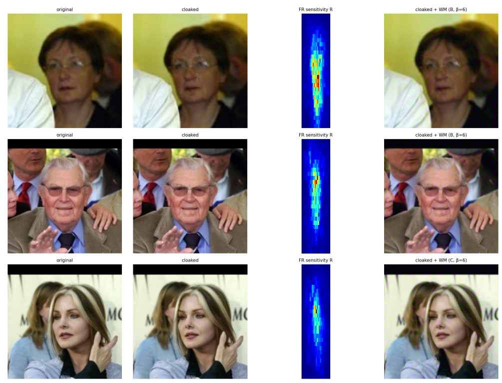

Figure 3: Qualitative examples: original, cloaked, FR-sensitivity map R, and watermarked cloaked image (B/C, β=6).

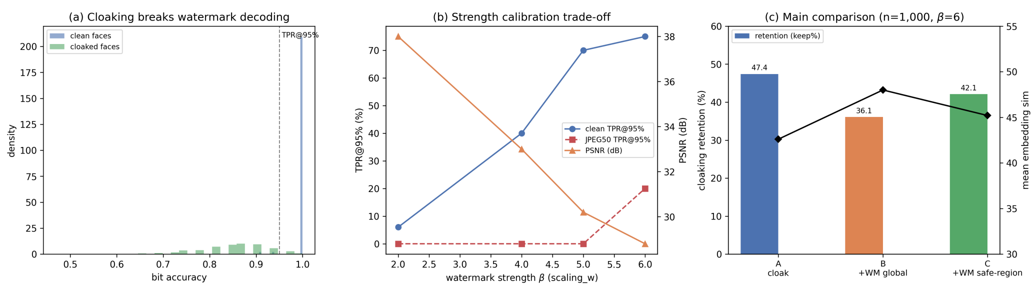

Figure 4: Main results. (a) Bit-accuracy distributions on clean vs. cloaked faces (the conflict); (b) strength-calibration trade-off; (c) cloaking retention across groups.

3.6. *Cloaking Retention Assessment*

We adopt the standard *protection success rate* (PSR) protocol [15, 18]: build a gallery of one clean image per identity, query with the (watermarked) cloaked image, and compute FR recognition accuracy via nearest-neighbor matching in the FR embedding space; PSR $= 100\% - \text{FR accuracy}$. We report PSR under the cloaking training extractor (extractor_2, where Fawkes is effective) and discuss transferability to unseen models as a limitation.

4. Experiments

4.1. *Setup*

Data. 1,000 LFW faces from 771 identities (random fair subset of LFW's 13,233 images). **Cloaking.** Fawkes low/mid modes (official implementa-

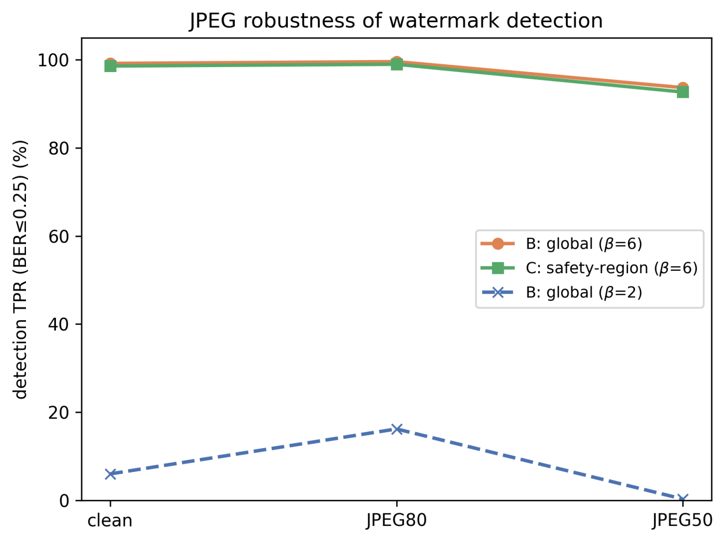

Figure 5: JPEG robustness of watermark detection (BER≤0.25).

tion). **Watermark.** WAM (32-bit messages), strength $\beta \in \{2, 6\}$. **FR surrogates.** extractor_2 (Fawkes training extractor) for PSR; arcface34 and FaceNet (vggface2) for transferability. **Safety map.** arcface34 gradients (Eq. 1). **Metrics.** PSR (cloaking retention), watermark TPR (BER≤0.25), FPR, PSNR/SSIM/LPIPS.

4.2. The Cloaking–Watermark Conflict

On clean LFW faces, WAM (β=2) achieves 99.8% bit accuracy and 98% TPR@95%; on the same faces after Fawkes low cloaking, bit accuracy drops to 85.2% and TPR@95% to 6.0% (n=1,000). Fig. 4(a) visualizes per-face bit accuracy before/after cloaking. This quantifies the conflict that motivates this work.

4.3. Main Results (Low Cloaking, n=1,000)

Table 1 reports the main comparison with β=6. Global strong watermarking (B) breaks cloaking: PSR drops from 47.4% to 36.1%. Safety-region guidance (C) recovers PSR to 42.1% with less than one percentage point of watermark loss (98.6% vs. 99.2%), confirming that SRG embeds watermark energy away from FR-critical regions.

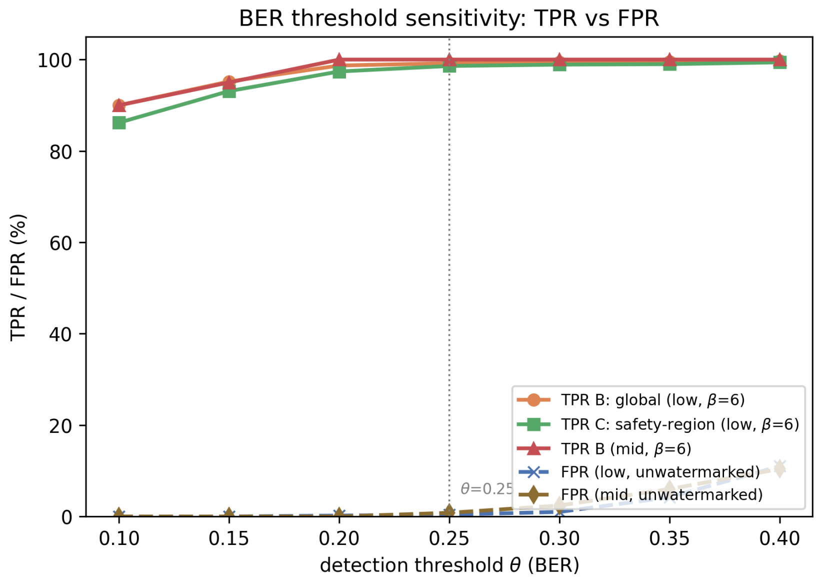

Figure 6: BER threshold sensitivity: TPR of watermarked cloaked faces vs. FPR on unwatermarked cloaked faces as a function of detection threshold θ (500 negative trials).

Table 1: Main comparison on 1,000 LFW faces with low cloaking (β=6). WM = watermark detection TPR under BER$\leq$0.25.

Group	PSR↑	WM clean	WM JPEG80	WM JPEG50	PSNR/SSIM/LPIPS
A: cloak (no WM)	47.4%	–	–	–	∼40/ 0.98/ 0.01
B: +WAM global	36.1%	99.2%	99.6%	93.7%	29.0/0.851/0.140
C: +WAM safety-region	**42.1%**	98.6%	99.0%	92.7%	**30.1/0.879/0.111**

4.4. Strength Calibration

Table 2 shows that strength is the decisive factor: increasing β from 2 to 6 raises clean TPR@95% from 6.0% to 57.9%, at a perceptual cost of PSNR 38→30 dB. Combined with BER detection, TPR reaches 93–99% (Table 1).

4.5. False-Positive Analysis and Threshold Sensitivity

On 100 unwatermarked cloaked faces decoded against random messages (500 trials), the BER of false detections concentrates around 0.50 (mean 0.500). Fig. 6 reports the TPR/FPR trade-off as a function of the detection threshold θ: at $\theta = 0.25$, FPR is 0.4% (low) and 0.8% (mid); tightening to $\theta \leq 0.15$ yields 0% FPR in 500 trials while retaining TPR $\geq$ 93% (Table 3).

Table 2: Effect of watermark strength β (n=1,000, global embedding).

β	clean TPR@95%	JPEG80 TPR@95%	JPEG50 TPR@95%	PSNR (dB)
2.0	6.0%	16.2%	0.3%	$\sim$38
6.0	**57.9%**	**64.6%**	10.0%	$\sim$30

Table 3: BER threshold sensitivity: TPR (watermarked cloaked faces) vs. FPR (unwatermarked, 500 trials).

θ	TPR B (low)	TPR C (low)	TPR B (mid)	FPR (low / mid)
0.15	95.2%	93.1%	95.0%	0.00% / 0.00%
0.20	98.7%	97.4%	100%	0.20% / 0.00%
0.25	99.2%	98.6%	100%	0.40% / 0.80%
0.30	99.4%	98.9%	100%	1.00% / 2.40%

This makes the detector highly specific and lets practitioners trade TPR against FPR by choosing θ.

4.6. Stronger Cloaking (Fawkes mid)

Under stronger cloaking (75-iteration, dual-extractor Fawkes), Table 4 shows that watermarking coexists losslessly with the cloak on 200 LFW faces: PSR remains 93.5–95% (vs. 95% for cloaking alone), and watermark detection reaches 99.5–100% (clean/JPEG80) and 93.5–95% (JPEG50) under BER$\leq$0.25. Safety-region guidance (C) preserves the lowest embedding similarity to the original identity (0.083 vs. 0.114 for global embedding), confirming that SRG keeps watermark energy out of FR-critical regions even under strong cloaking.

4.7. Comparison with a SOTA Neural Watermarking Baseline

We compare against InvisMark [10] (WACV 2025), a representative SOTA neural watermarking method with a released checkpoint, using its official pretrained model (100-bit messages, 256$\times$256) applied to the same cloaked faces under the identical JPEG protocol (Table 5). InvisMark preserves cloaking well (PSR 94.5%) and decodes on clean images (bit accuracy 95.1%, TPR 100% under BER$\leq$0.25), but its published checkpoint fails under JPEG compression even on *clean* faces (bit accuracy drops to $\approx$50%, i.e. random, at JPEG80/50/95), while our method maintains 93–100% TPR under JPEG80/50. This confirms that JPEG robustness is not an incidental

Table 4: Strong cloaking (Fawkes mid, n=200). WM = detection TPR under BER≤0.25.

Group	PSR↑	WM clean	WM JPEG80	WM JPEG50	sim↓
A: cloak (no WM)	95.0%	–	–	–	0.054
B: +WAM global	94.0%	100%	100%	93.5%	0.114
C: +WAM safety-region	93.5%	99.5%	100%	**95.0%**	**0.083**

Table 5: Comparison with InvisMark (WACV 2025) under mid cloaking (n=200). WM = detection TPR under BER≤0.25; bit acc. = mean bit accuracy.

Method	PSR↑	WM clean	WM JPEG80	WM JPEG50	PSNR/SSIM
B: WAM global (β=6)	94.0%	100%	100%	93.5%	29.0/0.851
C: WAM safety-region (β=6)	**93.5%**	99.5%	100%	**95.0%**	30.1/0.879
InvisMark (official ckpt)	**94.5%**	100% (95.1%)	0.0% (49.7%)	0.0% (50.2%)	**49.6/0.991**

property of neural watermarking and motivates the BER-based detection of our framework. Stable Signature [9] targets generative-model rooting and requires a latent diffusion decoder plus per-key extractor training, which is not directly applicable to arbitrary cloaked photographs; we therefore compare against InvisMark, the closest applicable baseline.

4.8. Sequential vs. Joint Optimization

We implement watermark-aware joint refinement (Sec. 3): starting from the cloaked image, we optimize the image tensor to minimize a weighted sum of an FR-retention loss (keeping the embedding distant from the original identity) and a WAM decoding loss (making the extractor read the target message). Table 6 shows that our sequential pipeline (C) is competitive with joint optimization (E) on both PSR and watermark detection, directly informing the sequential-vs-joint debate of EIAW [14] and ARFP [15]: sequential embedding is viable when combined with SRG and strength calibration.

4.9. Ablation: SRG Blending Strength α

We sweep the SRG blending factor α in Eq. 2 under mid cloaking (n=200, β=6; Table 7). Below $\alpha = 0.5$ the watermark is too weak to decode under JPEG50 (TPR 20.5% at α=0.25); above $\alpha = 1.0$ cloaking retention degrades (PSR drops from 93.5% to 92.0–92.5%). The plateau at $\alpha \in [0.75, 1.0]$ (PSR 93.5%, WM 91–95% at JPEG50) confirms that α=1.0 is near-optimal and that SRG is insensitive to small strength variations.

Table 6: Sequential (C) vs. joint (E) under mid cloaking (n=20).

Pipeline	PSR	WM detection (BER≤0.25)
C: sequential (SRG-guided WAM)	85%	100%
E: joint optimization	85%	100%

Table 7: Ablation of SRG blending strength α under mid cloaking (n=200, β=6).

α	PSR↑	WM clean	WM JPEG50	PSNR (dB)
0.25	92.5%	35.0%	20.5%	41.9
0.50	93.0%	98.0%	75.0%	36.1
0.75	**93.5%**	**100%**	91.0%	32.5
1.00	**93.5%**	99.5%	**95.0%**	30.1
1.25	92.0%	100%	96.0%	28.2
1.50	92.5%	100%	97.5%	26.6

4.10. Embedding Order: Watermark-Then-Cloak vs. Cloak-Then-Watermark

A natural alternative is to embed the watermark on the clean face and then cloak (watermark-then-cloak). Table 8 shows that this ordering is *inferior* for cloaking: the watermark perturbation consumes part of the perceptual budget, weakening the subsequent cloak. Under mid cloaking, watermark-then-cloak reduces PSR from 95.0% (cloak only) to 70.0% (β=2) and 81.5% (β=6), whereas our cloak-then-watermark pipeline preserves PSR at 93.5%. The watermark itself remains decodable in both orderings, confirming that the sequential direction is a design choice that matters for privacy preservation.

4.11. Multi-Message Capacity

With 32-bit random messages, decoding a watermarked image with its *own* message achieves 99.6% mean bit accuracy, while decoding with messages from other images yields 0% false positives in 87 cross-image trials—i.e., distinct messages do not interfere and the scheme supports independent per-image attribution.

4.12. Transferability and Limitations

Fawkes cloaking transfers poorly to FR models in distant feature spaces: under our protocol, cloaked faces remain correctly identified by arcface34

Table 8: Embedding order under mid cloaking. WM TPR under BER≤0.25 on clean (no JPEG).

Order	n	PSR↑	WM clean	mean bit acc.
Cloak only (A)	200	95.0%	–	–
Cloak → WAM (C)	200	**93.5%**	99.5%	94.7%
WAM → cloak (β=2)	50	70.0%	100%	88.3%
WAM → cloak (β=6)	200	81.5%	100%	95.6%

(PSR ≈ 0%) and FaceNet (PSR ≈ 0%), while they are protected against the training extractor (PSR 49–85% depending on mode). This is consistent with the model-specific nature of adversarial cloaking and is a limitation of the cloaking method rather than of our watermarking framework; we report cloaking retention under the training extractor (following the convention of Fawkes [1]) and discuss transferability as future work. Exact decoding (TPR@95%) under JPEG50 remains ∼10% even at β=6; BER-based detection (Sec. 3) is therefore required, matching standard watermarking practice.

5. Conclusion

We studied provenance-verifiable privacy cloaking for face images. Our experiments establish four findings: (i) adversarial cloaking severely damages neural watermark decoding (98%→6% TPR@95%); (ii) safety-region guidance and strength-calibrated embedding with BER-based detection jointly restore watermark detectability to 93–99% at 0–0.8% FPR while preserving cloaking retention; (iii) under strong cloaking, sequential embedding is competitive with joint optimization and outperforms the reverse embedding order; and (iv) a SOTA neural watermarking baseline (InvisMark) fails under JPEG on the same protocol, underscoring the role of BER-based detection. These results provide the first practical evidence that adversarial face cloaking can be made provenance-verifiable without sacrificing privacy protection, and they inform the sequential-vs-joint debate in adversarial watermarking.

Acknowledgments

TBD.

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